Fossil Fuel Prices and Capital Cost

A Machine Learning Driven Study on Energy Transition

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Presented by Shilin Ou

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Financing conditions matter for the energy transition

Wind and solar concentrate spending at the start of a project.

- Debt and equity providers require a return on the capital they commit.
- A higher financing cost raises the revenue a project must earn.
- Regional financing differences can change modeled transition pathways.

WACC

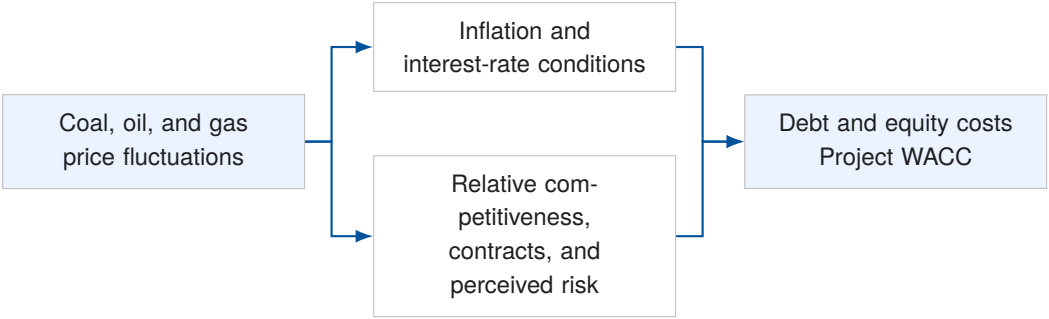
Weighted average cost of capital

$$\underbrace{Lr_D(1-\tau)}_{\text{debt}} + \underbrace{(1-L)r_E}_{\text{equity}}$$

Ameli et al., Nature Communications (2021); Calcaterra et al., Nature Energy (2024).

One price signal, several financing channels

What links fossil fuel prices to energy-project WACC?



We compare these signals across China, Europe, and the Middle East.

Mechanisms from Melodia and Karlsson (2022), Lieberman and Doherty (2007), and Duma and Muñoz Cabré (2023).

Data connect historical signals with future WACC scenarios

Input	Content	Role
Fuel prices	Monthly coal, oil, and natural gas series	Price-forecast inputs
Macroeconomy	Inflation, growth, unemployment, interest-rate and monetary indicators	Regional inflation forecasts
WACC	Calcaterra et al. model projections, aligned to 2026–2035	Financing scenarios

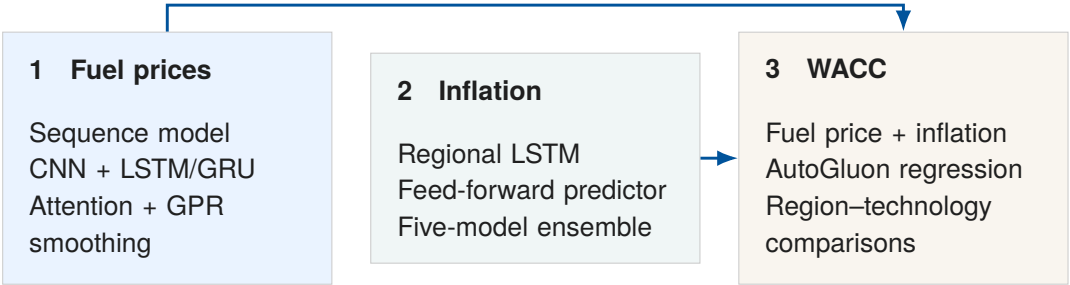
3 regions China, Europe, Middle East

120 monthly rows per cleaned region

9 technologies wind, solar, hydro, biomass, nuclear, CCS, coal, and gas

WACC values are modeled scenarios, rather than future realized financing observations. Coal and gas generation serve as conventional-technology comparisons.

A three-stage machine-learning framework



Comparison structure 3 regions × 3 fuels × 9 technologies = 81 combinations

The region, fuel, and technology axes organize the comparisons. Separate chronological diagnostics examine the stability of fitted relationships.

Fuel-price forecasting combines temporal learning and smoothing

Sequence model

- A 12-month input window captures recent price dynamics.
- Seasonal sine/cosine features represent the month of year.
- CNN, recurrent layers, and attention learn temporal representations.

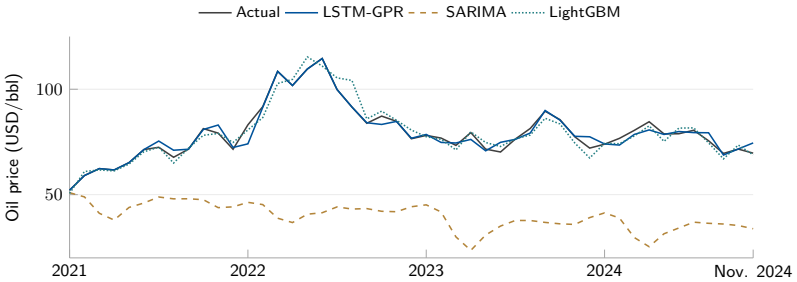
Gaussian process

- An RBF kernel smooths the neural forecast path.
- The resulting path feeds the WACC comparison.

Its conditional variance does not represent the full uncertainty of the neural forecasting stage.

Lindemann et al. (2021); Lahmiri (2024); Rasmussen (2004). Implementation settings are provided in the paper appendix.

The archived oil-price comparison favors LSTM-GPR



0.963
R^2

1.86%
MAPE

2.416
RMSE (USD/bbl)

47 observations in the archived 2021–2024 workbook. Metrics were recomputed from saved outputs; separate fixed-origin baselines are retained in the repository.

WACC regressions and chronological checks serve different roles

Original AutoGluon fits

$$\mathrm{WACC}_{r,p,t} = f_{r,p,e}(P_{e,t}, \pi_{r,t})$$

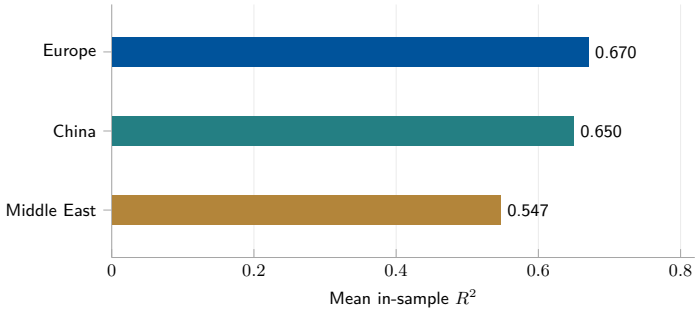
- One fuel price and an inflation covariate
- Bagging and stacking
- In-sample R^2, MAPE, and RMSE

Additional diagnostics

- OLS/Ridge with cleaned regional inputs
- Train 2026–2032; test 2033–2035
- Rolling one-year-ahead checks
- Compare fuel + inflation + trend against a time trend

Fit describes the modeled relationship. Chronological tests measure how that relationship transfers to later scenario years.

Regional fit differs across the original specifications



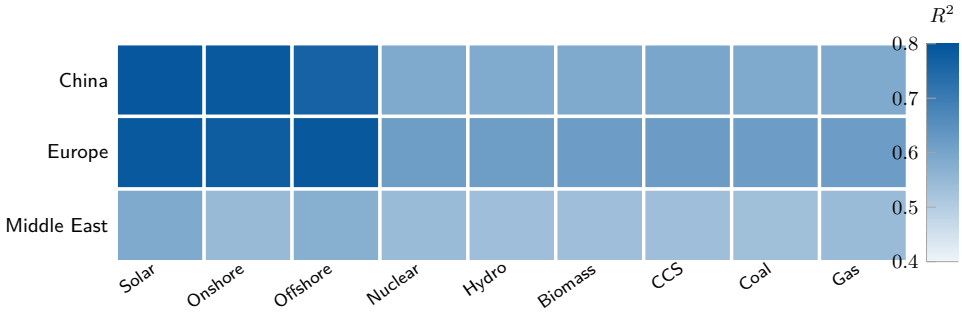
Europe and China show stronger fit.

Middle Eastern specifications have a lower average fit and larger percentage errors.

These comparisons describe joint fuel-price and inflation fits. R^2 does not measure the magnitude of a price shock.

Recalculated regional means from the archived 81-row AutoGluon results workbook.

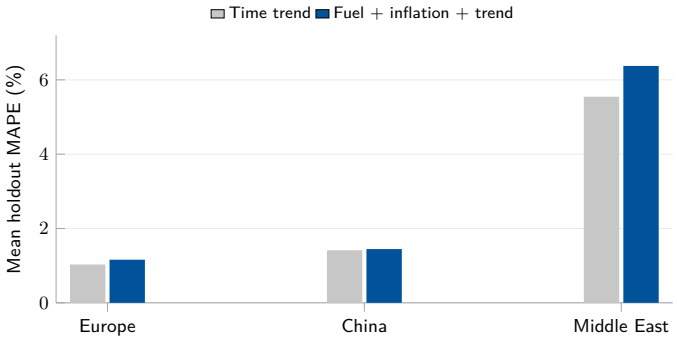
Wind and solar have the strongest average fits



Solar 0.718 **Offshore wind** 0.706 **Onshore wind** 0.700

Technology means across three fuels and three regions; heatmap cells average across fuels. Nuclear, hydro, biomass, CCS, coal, and gas are around 0.58 overall.

Later-year checks show limited gains beyond a time trend



Additional information varies by specification.

Positive incremental out-of-sample R^2:

China	8/27
Europe	6/27
Middle East	6/27

Small MAPE can coexist with negative R^2 when the test-period WACC range is narrow.

Ridge diagnostics on cleaned scenario panels; fixed holdout: 2033–2035. Counts compare fuel + inflation + trend against time trend alone.

Historical diagnostics identify selected macroeconomic links

Target	Fuel	Lag	p-value
China real interest rate	Coal	1 year	0.00018
China real interest rate	Coal	2 years	0.00691
China real interest rate	Gas	1 year	0.01915
Europe inflation	Gas	2 years	0.00952

Energy-price changes can contain information about later macro conditions.

Selected exploratory Granger-style tests on annual historical data. Targets are first-differenced and prices log-differenced; displayed p-values are unadjusted. These are transmission diagnostics.

Regional institutions guide the next financing scenarios

Region	Relevant context	Financing question
Europe	Imported energy and policy credibility	How can predictable contracts and risk transfer support investment?
China	International spillovers and domestic price regulation	How do public finance and price rules alter exposure?
Middle East	Oil dependence and domestic energy-pricing institutions	How can financing capacity support diversification?

Polzin et al. (2019); He and Lin (2017); Karanfil and Omgba (2023); Fattouh and El-Katiri (2013). The paper motivates these scenarios rather than estimating their policy effects.

Selected references

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The complete checked bibliography is provided with the paper and editable source package.

Authorship, author notes, and acknowledgments

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Financing comparisons benefit from regional detail

Three takeaways

1. A reproducible pipeline connects price forecasts, inflation, and WACC scenarios.
2. Fit varies across regions and technologies; wind and solar fit most closely.
3. Time-split tests show where fuel signals add information beyond a trend.

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Code + materials

Dataset snapshot

Corrected diagnostics and editable LaTeX sources are in the GitHub repository.